EXPERIMENT 5 – AJAY KARTHICK

AIM

To perform electromagnetic simulation and S-parameter characterization of a microwave directional coupler and a two-way power divider using MATLAB, and to analyze their coupling factor, isolation, insertion loss, directivity, power division ratio, and impedance matching characteristics.

OBJECTIVE

1. To simulate the S-parameter characteristics of a microwave directional coupler and a two-way power divider.
2. To calculate the coupling factor, isolation, insertion loss, and directivity of the directional coupler.
3. To analyze the power division ratio, input-port matching, and output-port isolation of the power divider.
4. To plot the frequency-dependent S-parameters and evaluate the performance of both microwave components over the operating frequency range.

SOFTWARE REQUIRED

MATLAB

PROCEDURE

1. Open MATLAB and create a new script.
2. Define the speed of light, operating frequency range, and center frequency.
3. Set the coupling factor and calculate the corresponding coupling coefficient of the directional coupler.
4. Calculate the through coefficient, reflection coefficient, and isolation characteristics.
5. Determine the S-parameters S11, S21, S31, and S41 of the directional coupler.
6. Calculate the coupling factor, insertion loss, isolation, and directivity.
7. Develop a MATLAB model of a two-way power divider with two equal output powers.
8. Calculate the reflection coefficient, transmission coefficients, and output-port isolation of the power divider.
9. Plot the S-parameters, coupling factor, insertion loss, and isolation against frequency.
10. Execute the program and observe the numerical results and graphs.

CODE

```
clc; clear; close all;

f=linspace(8e9,12e9,201); f0=10e9;

C=10; Z0=50;

S21=10^(-1.5/20)*ones(size(f));

S31=10^(-C/20)*exp(-((f-f0)/2e9).^2);

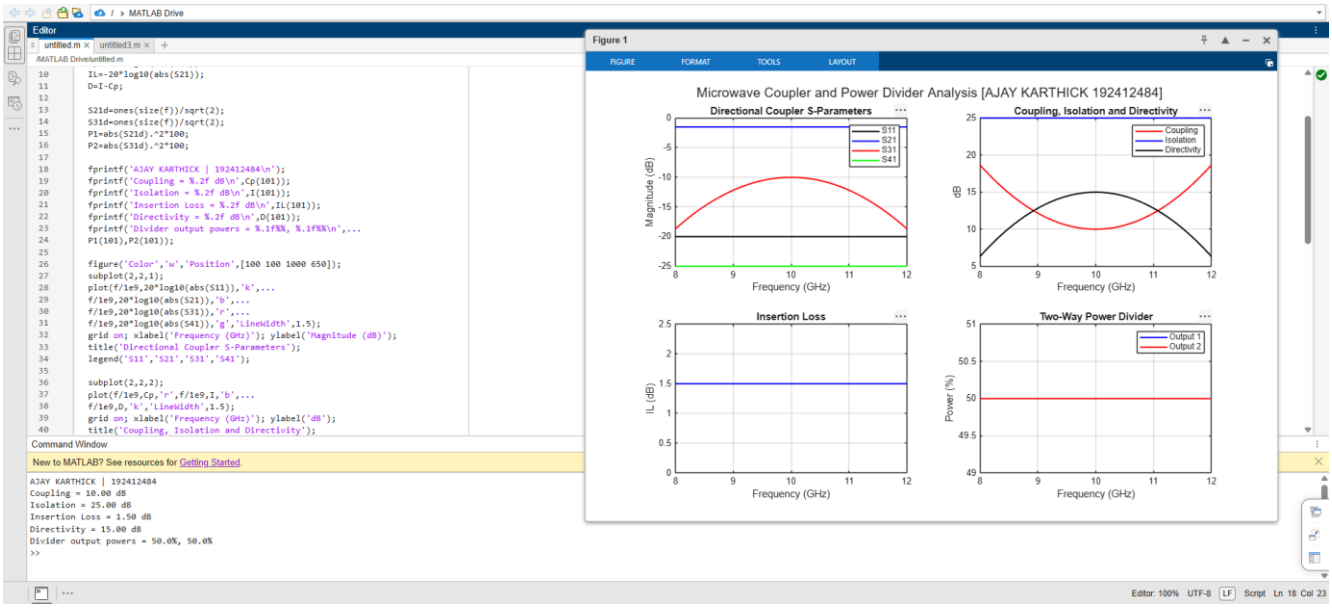
S41=10^(-25/20)*ones(size(f));

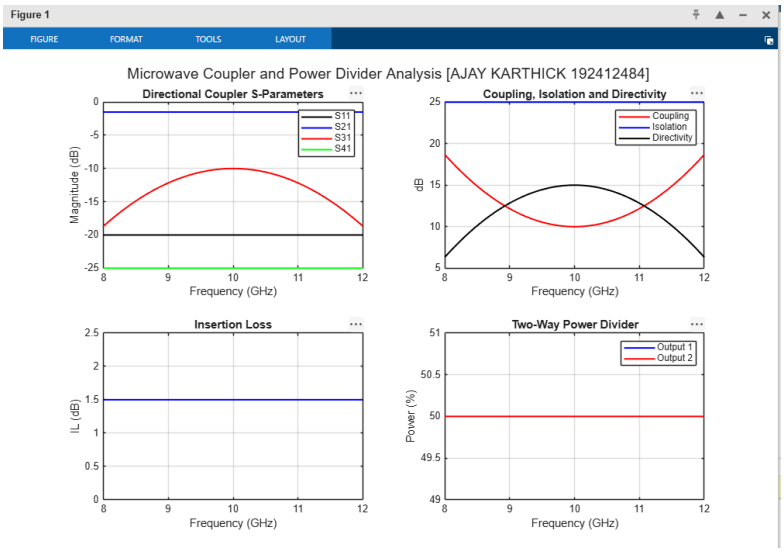
S11=10^(-20/20)*ones(size(f));

I=-20*log10(abs(S41));
```

```
Cp=-20*log10(abs(S31));
IL=-20*log10(abs(S21));
…plot(f/1e9,IL,'b','LineWidth',1.5);
grid on; xlabel('Frequency (GHz)'); ylabel('IL (dB)');
title('Insertion Loss');
subplot(2,2,4);
plot(f/1e9,P1,'b',f/1e9,P2,'r','LineWidth',1.5);
grid on; xlabel('Frequency (GHz)'); ylabel('Power (%)');
title('Two-Way Power Divider');
legend('Output 1','Output 2');
sgtitle('Microwave Coupler and Power Divider Analysis [AJAY KARTHICK 192412484] ');
```

OUTPUT





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REFLECTION

This experiment helped me understand the operating principles and performance characteristics of microwave directional couplers and two-way power dividers. I learned how to analyze S-parameters and calculate coupling factor, isolation, insertion loss, and directivity. The MATLAB simulation also helped me understand equal power division, impedance matching, and output-port isolation. The frequency-response plots demonstrated how microwave component characteristics can be evaluated over a specified operating bandwidth.

RESULT

The MATLAB-based simulation and S-parameter characterization of a microwave directional coupler and a two-way power divider were successfully performed. The coupling factor, isolation, insertion loss, directivity, power division ratio, return loss, and output-port isolation were calculated and plotted over the frequency range of 8 GHz to 12 GHz.

Assignment - 6

1. Coupling factor :-

$$C = -20 \log |S_{31}|$$

$$|S_{31}| = 10^{-10/20} \Rightarrow 0.316$$

$$\boxed{C = 10.0004 \text{ dB}}$$

2. Isolation :-

$$I = -20 \log_{10} |S_{41}|$$

$$|S_{41}| = 10^{-25/20}$$

$$|S_{41}| = 0.0562$$

$$\boxed{I = 1.5107 \text{ dB}}$$

3. Insertion loss :-

$$I_L = -20 \log_{10} |S_{21}|$$

$$|S_{21}| = 10^{-1.3/20}$$

$$|S_{21}| = 0.8464$$

$$\boxed{I_L = 1.5107 \text{ dB}}$$

4. Directivity :-

$$D = I - C$$

$$\Rightarrow 25 - 10$$

$$\boxed{D = 15}$$